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QUARTERLY EXAMINATION - 2025

CLASS : 12 Reg.No

Time : 3.00 Hours **CHEMISTRY** Total Marks : 70

PART - A

I Answer all the questions. **15x1=15**

1. The metal extracted by Hall-Heroult process is
   a) Cu b) Al c) Zn d) Ni
2. Carbon atoms in Fullerene with formula C_{60} have ..... hybridisation.
   a) sp^2 b) sp^3 c) sp^3d d) sp
3. Which of the following is used as the source of gamma rays?
   a) Xe b) Ar c) Rn d) Kr
4. Which of the following Lanthanoid ions is diamagnetic?
   a) Eu^{2+} b) Yb^{2+} c) Ce^{2+} d) Sm^{2+}
5. CsCl has bcc arrangement, its unit cell edge length is 400 pm, its inter atomic distance is
   a) 400 pm b) 800 pm c) $\sqrt{3}\times 100$ pm d) $\left[\frac{\sqrt{3}}{2}\right]\times 400$ pm
6. In a first order reaction $x \rightarrow y$: if k is the rate constant any the initial concentration of the reactant x is 0.1M, then, the half life is
   a) $\left[\frac{\log 2}{k}\right]$ b) $\left[\frac{0.693}{(0.1)k}\right]$ c) $\left[\frac{ln2}{k}\right]$ d) None of these
7. On hydrolysis, PCl_3 gives
   a) H_3PO_3 b) PH_3 c) H_3PO_4 d) $POCl_3$
8. pH of a saturated solution of $Ca(OH)_2$ is 9. The solubility product (K_{sp}) of $Ca(OH)_2$
   a) 0.5×10^{-15} b) 0.25×10^{-10} c) 0.125×10^{-15} d) 0.5×10^{-10}
9. Which of the following compounds on reaction with methyl magnesium bromide will give tertiary alcohol.
   a) Benzaldehyde b) Porpanoic acid c) Methyl Propanoate d) Acetaldehyde
10. $CH_2 = CH_2 \xrightarrow[ii)\ Zn/H_2O]{i)\ O_3} X \xrightarrow{NH_3} y$ 'y' is
    a) Formaldehyde b) di acetone ammonia c) Hexamethylene tetra amine d) Oxime
11. The number of moles of acidified $KMnO_4$ required to oxidize 1 mole of ferrous oxalate ($Fe C_2O_4$) is
    a) 5 b) 3 c) 0.6 d) 1.5
12. At 25 C, ionic product constant K_w of water is 1.00×10^{-14} . Its value at 40^0C is
    a) 1.00×10^{-14} b) 1.114×10^{-15} c) 2.71×10^{-14} d) 2.95×10^{-15}
13. Assertion : The C - O - C bond angle in ether is slightly greater than the tetrahedral bond angle.
    Reason : Repulsive interaction between the two bulkier alkyl groups.
    a) Assertion is true but Reason is false
    b) Both Assertion and Reason are true and Reason is the correct explanation of Assertion
    c) Both Assertion and Reason are false
    d) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion
14. Match items in Column - I with the items Colum - II and assign the correct code

Column - I	Column - II		A	B	C	D
A) Tollens Reagent	(i) N_2H_4 / C_2H_5ONa	a)	(i)	(ii)	(iii)	(iv)
B) Octyl acetate	(ii) Ammonical silver nitrate	b)	(ii)	(iii)	(iv)	(i)
C) Acetic acid	(iii) Orange	c)	(iii)	(iv)	(i)	(ii)
D) Wolf Kishner Reduction	(iv) Table Vinegar	d)	(iv)	(iii)	(ii)	(i)

15. The most common oxidation state of actinoids is
    a) +2 b) +3 c) +4 d) +6

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PART - B

II Answer any six questions (Question No:24 is compulsory) **6x2=12**

16. What are the various steps involved in extraction of pure metal from their ores.
17. CO is a reducing agent. Justify with an example.
18. Give a reaction between nitric acid and basic oxide.
19. Why 'd' block elements exhibit variable oxidation state?
20. Define Packing efficiency?
21. Define Half life period.
22. What are Lewis acids and bases? Give one example for each.
23. Give the uses of Diethyl ether.
24. Show that in case first order reaction the time required for the completion of 99% is twice the time required for the completion of 90% of the reaction.

PART - C

III Answer any six questions (Question No:33 is compulsory) **6x3=18**

25. Write the difference between mineral and ore.
26. Write note on Fisher tropsch synthesis.
27. Why HF not stored in glass bottle?
28. What is Lanthanide contraction? and what are the effect of Lanthanide contraction.
29. Calculate the number of atoms in a fcc unit cell.
30. Derive integrated rate law for a zero order reaction.
31. Explain common ion effect with suitable example.
32. Write the preparation of Urotropine and give the uses.
33. Write the test for Secondary alcohol with equation.

PART - D

IV Answer the following questions. **5x5=25**

34. a) Explain Zone refining process. **[OR]** (5)

    b) i) Find the oxidation state of Halogen in the following compounds.

    (1) OF_2 (2) O_2F_2 (2)

    ii) Complete the following reactions.

    1) $B(OH)_3+NH_3$ -------- ?

    2) $Ca(OH)_2+Cl_2$ ----------? (3)

35. (a) i) Write short notes on Holmes signal. (3)

    ii) How will you prepare chlorine from Deacon's process. **[OR]** (2)

    (b) What are interstitial compounds? Write the properties of interstitial compound. (5)

36. (a) Differentiate crystalline solids and amorphous solids **[OR]** (5)

    (b) i) Explain the effect of Catalyst on reaction rate with an example. (3)

    ii) Define rate law. (2)

37. (a) Derive an expression for Ostwald's Dilution Law. **[OR]** (5)

    (b) Compound (A) molecular formula C_6H_6O gives purple coloration with neutral $FeCl_3$. (5)
    Compound (A) reacts with benzene diazonium chloride to give compound (B) and it also react with Nitrating mixture at 298k to give compound (c). Identify the compounds A, B, and C and write down the equations.

38. (a) i) How malachite green is prepared from Benzaldehyde? (3)

    ii) Explain Rosenmund reduction. **[OR]** (2)

    (b) Explain reducing nature of formic acid with suitable example. (5)

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